

Tutorial Quiz-10 (Max. Duration: 20 mins)

Instructions: Please show your work clearly and specify your assumptions (if any).

You are designing the logic for a children's electronic board game. The game uses a digital 6-sided die that outputs a 3-bit binary number (D_2 , D_1 , D_0). A roll of 1 is 001, a roll of 6 is 110.

The game has a special rule: If a player rolls an Even Number (2, 4, or 6), a green LED lights up to grant them a Bonus Turn (Bonus = 1). If they roll an odd number, the LED stays off (Bonus = 0).

Q1. (2 points) Create the truth table for the Bonus LED logic for all 8 possible 3-bit inputs.

Q2. (2 points) Write the un-simplified SOP logic expression for Bonus LED logic.

Q3. (4 points) Use a 3-variable K-map to find the absolute simplest Boolean expression for the Bonus LED. [Hint: Pay attention to the **Don't care** states]

Q4. (2 points) Draw the Bonus LED logic circuit for your simplified expression using any necessary combination of NOT, 2-input AND, and 2-input OR gates (you are not required to use all three types).

		$D_1 D_0$			
D_2		00	01	11	10
	0				
	1				

Quiz 10 Solution

① Truth table :-

D_2	D_1	D_0	O/P (Bonus)	
0	0	0	X	
0	0	1	0	$\Rightarrow$ ②
0	1	0	1	
0	1	1	0	$\Rightarrow$ Bonus(B)=1 for
1	0	0	1	even numbers i.e.
1	0	1	0	2, 4, 6
1	1	0	1	$\Rightarrow$ Bonus(B)=0 for
1	1	0	X	odd numbers i.e.
				1, 3, 5

② Unsimplified SOP expression
Bonus = 1 for minterms

② $B = \bar{D}_2 D_1 \bar{D}_0 + D_2 \bar{D}_1 \bar{D}_0 + D_2 D_1 \bar{D}_0$

③

② $\Rightarrow$

	$D_1 D_0$	00	01	11	10
D_2	0	X			1
	1	1		X	1

* 4 corner cells form quad

$B = \bar{D}_0$

②

Alternate solution if only 2 pairs considered

D_1, D_0		00	01	11	10
D_2	0	X			1
	1	1		X	1

← Pair (points to the pair (0,1) in column 10)

← Pair (points to the pair (0,1) in column 00)

← 0.5

$$B = \bar{D}_1 \bar{D}_0 + D_1 \bar{D}_0 \quad \leftarrow \textcircled{1}$$

$$= \bar{D}_0 (\bar{D}_1 + D_1)$$

$$B = \bar{D}_0 \quad \leftarrow \textcircled{1} \quad [\bar{D}_1 + D_1 = 1]$$

Alternate solution if one pair considered

D_1, D_0		00	01	11	10
D_2	0				1
	1	1			1

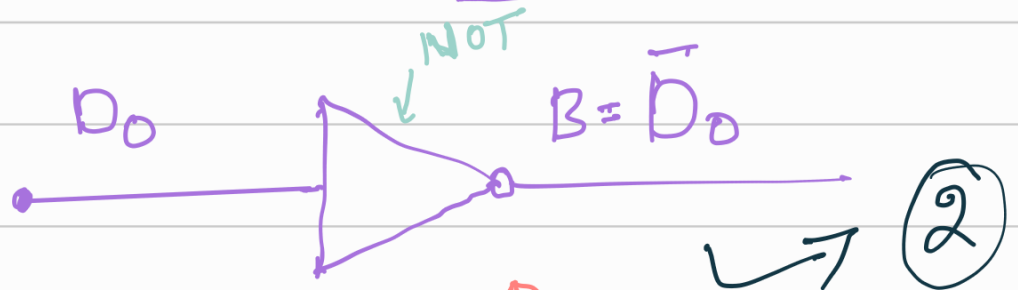
← one pair (points to the pair (0,1) in column 10)

← 0.25

$$B = D_2 \bar{D}_1 \bar{D}_0 + D_1 \bar{D}_0 \quad \leftarrow \textcircled{0.25}$$

④ for first solution of 3rd part where quad considered

Logic circuit : $B = \bar{D}_0$



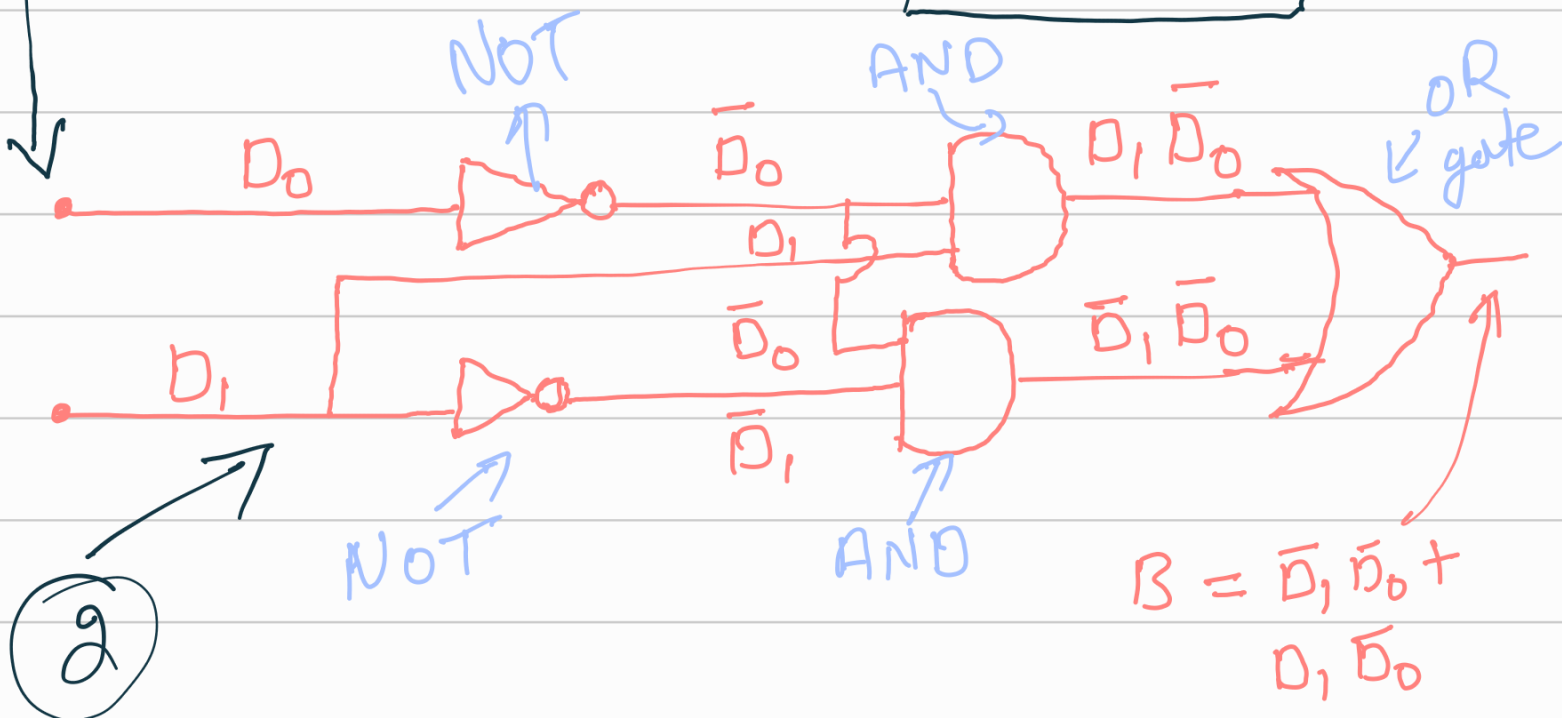
for second solution : -

→ without simplification

$$B = \bar{D}_1 \bar{D}_0 + D_1 \bar{D}_0$$

if simplified

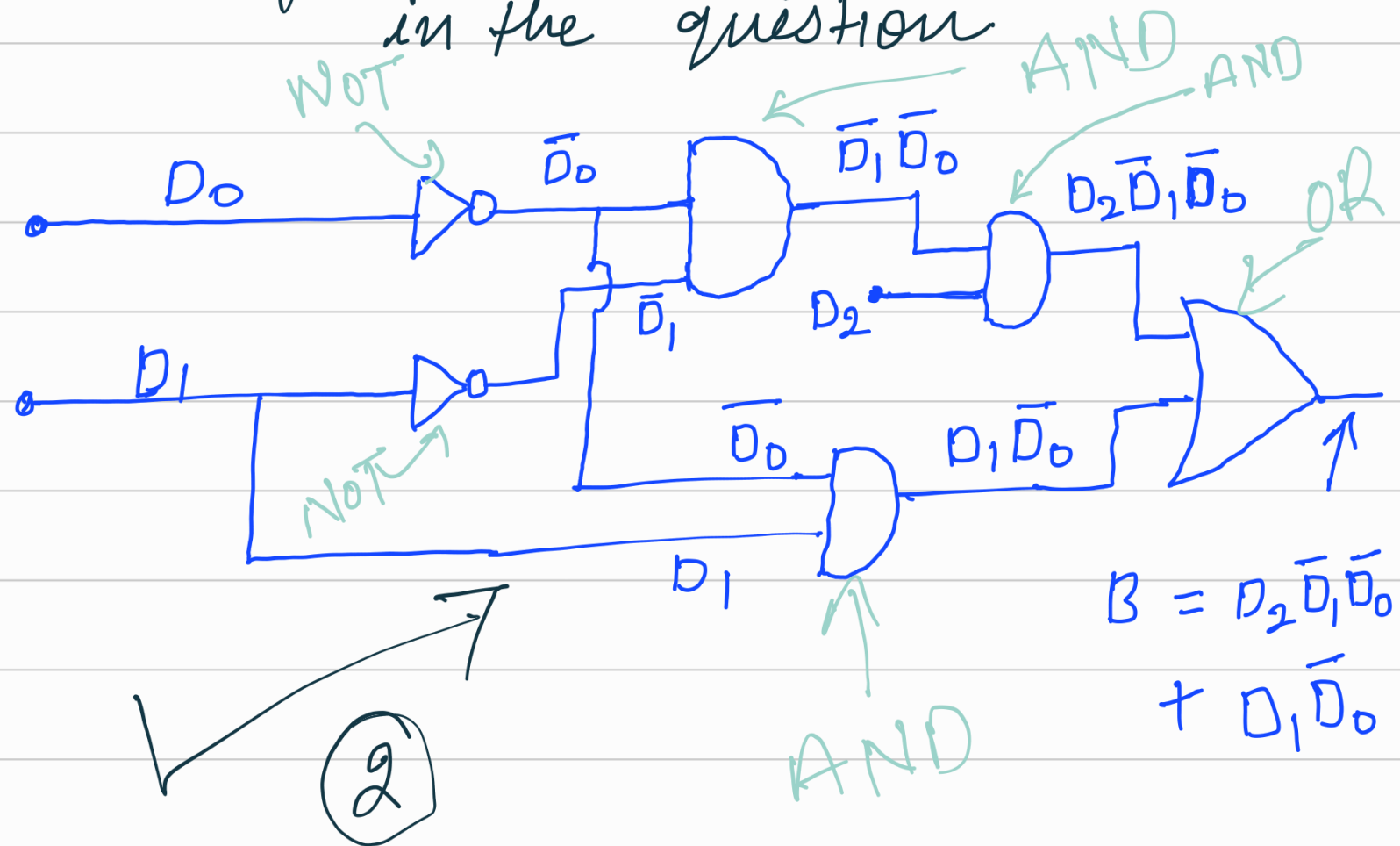
$$B = \bar{D}_0$$



for 3rd possible solution:-

$$B = D_2 \bar{D}_1 \bar{D}_0 + D_1 \bar{D}_0$$

only 2-input gates are mentioned in the question

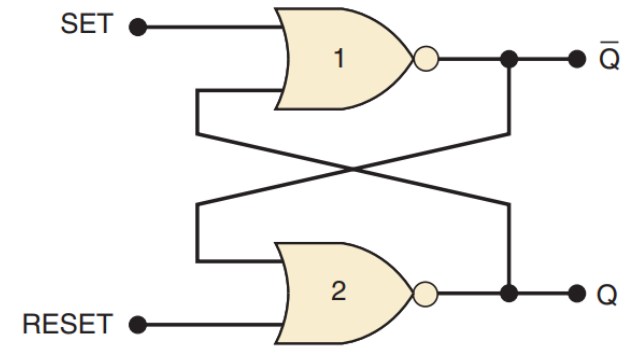


Tutorial Quiz-11 (Max. Duration: 20 mins)

Instructions: Please show your work clearly and specify your assumptions (if any).

You are designing a water pump controller using an Active-High SR Latch (i.e. $Q = 1$ turns pump ON). Inputs to the system:

- $L_LOW = 1$ (Tank is empty) – Sensor at the bottom of the tank
- $L_HIGH = 1$ (Tank is full) – Sensor at the top of the tank
- $EN = 1$ (Master switch enabled)
- $E_STOP = 1$ (Emergency stop pressed)



Conditions for determining S and R:

- 1) Pump ON: Triggers when the tank is empty AND the master switch is enabled.
- 2) Pump OFF: Triggers when the tank is full OR the emergency stop is pressed.

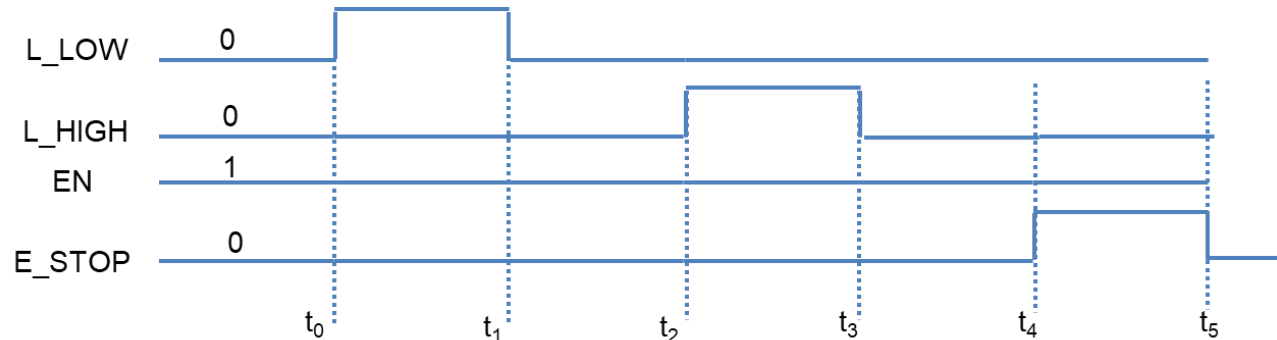
Q1. (4 points) Write the Boolean logic equations for the latch inputs (S and R).

Q2. (2 points) Sketch the circuit diagram connecting your logic gates to the SR Latch block (use the symbol of the SR latch).

For the given events (waveform):

Q3. (2 points) Draw the internal S and R waveforms based on their logic equations.

Q4. (2 points) Draw the final Q (Pump) waveform, identifying where the "Set," "Reset," and "Hold" states occur. Assume $Q=0$ before t_0 .



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- $EN = 1$ (Master switch enabled)
- $E_STOP = 1$ (Emergency stop pressed)

Conditions for determining S and R:

- 1) Pump ON: Triggers when the tank is empty AND the master switch is enabled.
- 2) Pump OFF: Triggers when the tank is full OR the emergency stop is pressed.

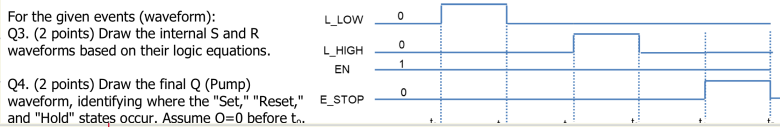
Q1. (4 points) Write the Boolean logic equations for the latch inputs (S and R).

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Q1

PUMP ON ($Q = 1$)

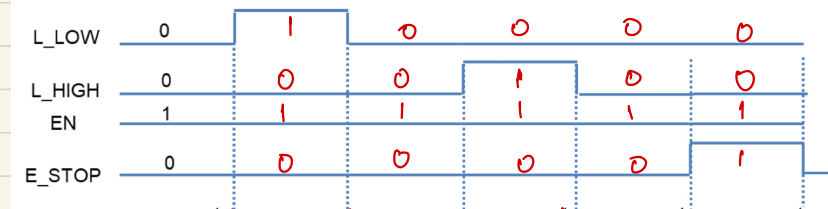
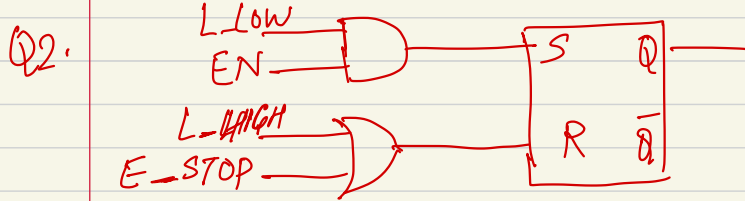
SET condition

$$S = L_LOW \cdot EN$$

PUMP OFF ($Q = 0$)

RESET condition

$$R = L_HIGH + E_STOP$$



Q4

S	R	Q_{n+1}	$\bar{Q}$
0	0	Q_n	
0	1	0	
1	0	1	
1	1	invalid	

Q3



Tutorial Quiz-12 (Max. Duration: 25 mins)

Instructions: This quiz uses negative marking for questions to discourage guessing. Unanswered questions receive 0 points. Read the point values carefully in the Google form.

You are analyzing a custom 3-bit Synchronous Sequential Circuit built using three D-Flip Flops (Q_2 , Q_1 , Q_0).

Look at the **last three digits** of your official Roll Number. Convert them into your initial 3-bit state (Q_2 Q_1 , Q_0) at Clock Cycle 0 using this rule:

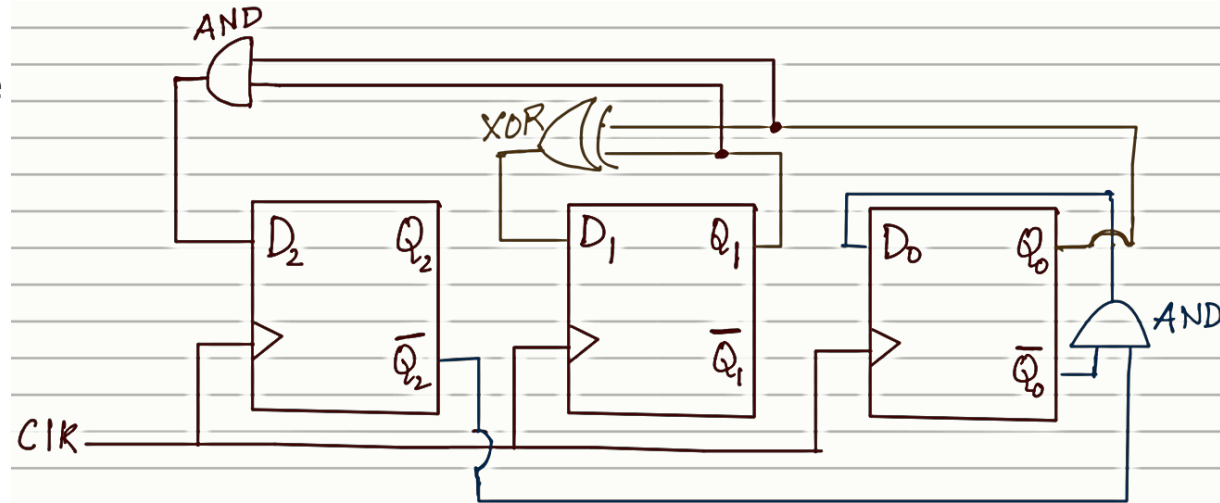
- Even digit (0, 2, 4, 6, 8) = 0
- Odd digit (1, 3, 5, 7, 9) = 1

Q1. (1 Mark) What is your 3-bit starting state?

At Clock Cycle 0, the Flip-Flops are externally preset to your personal 3-bit starting state (determined in Question 1). From Clock Cycle 1 onwards, the circuit runs freely based on the clock edges. Use your scratch paper to trace your specific state transitions.

Q3. (4 Marks) Trace your circuit's state transitions carefully. Starting from your initial state at Clock Cycle 0, **how many total rising clock edges** will it take for the circuit to arrive at state 000? (If Initial state is 000, count the edges for one full loop)

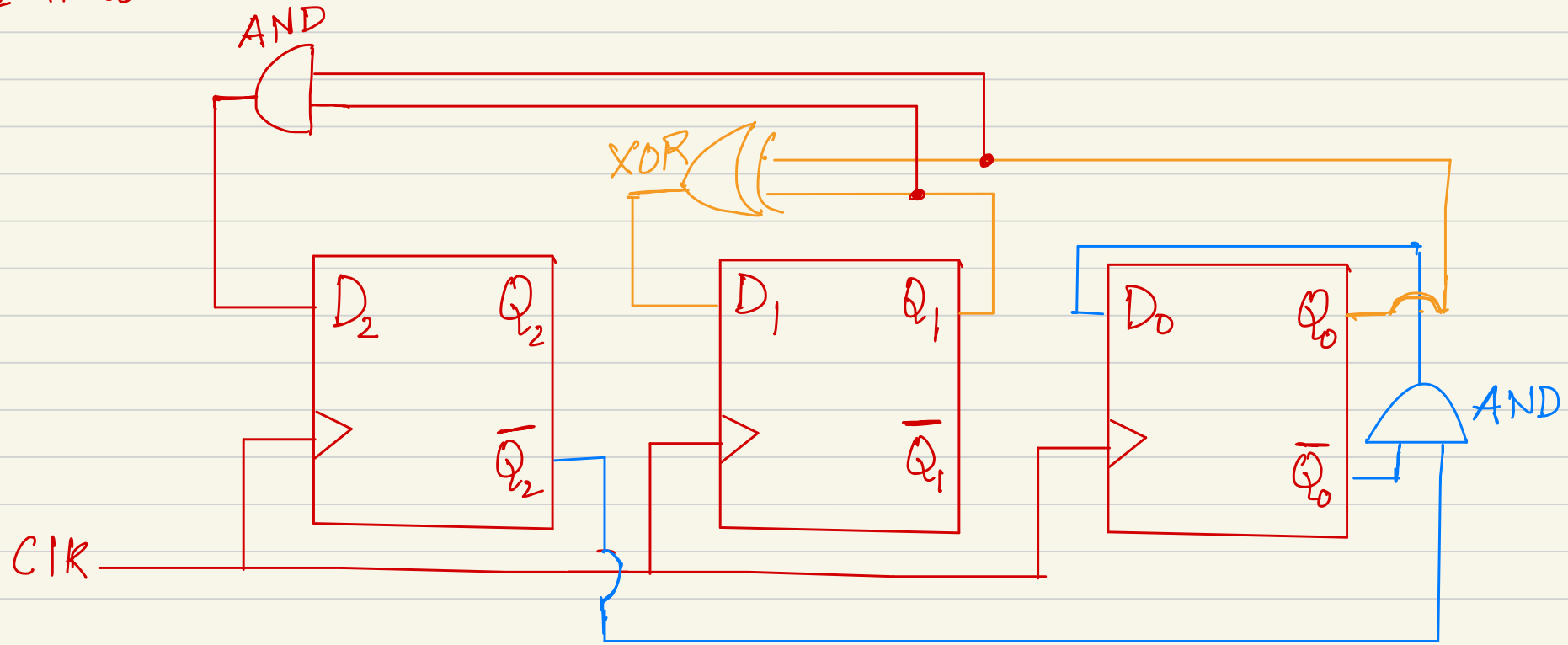
Q4. (2 Marks) Starting immediately after 000 (Q3), **what are the next states** the circuit visits before looping back to zero?



Q2. (3 Marks) For the given logic circuit, after the very first rising clock edge, **what is the exact 3-bit state of the circuit at the end of Clock Cycle 1?**

Tutorial Quiz 12

1) Assuming last three digits of my Roll no. to be 999, which means $Q_2 Q_1 Q_0 = 111$.



$$D_0 = \overline{Q_2} \cdot \overline{Q_0}, \quad D_1 = Q_1 \oplus Q_0, \quad D_2 = Q_1 \cdot Q_0$$

2). If 111 is loaded before clock 1 edge, $\Rightarrow D_2 = 1, D_1 = 0, D_0 = 0$
 $\Rightarrow$ At clock edge 1: $Q_2 Q_1 Q_0 = 100$ [D passes to output].

3) at clock edge

	Q_2	Q_1	Q_0	D_2	D_1	D_0
0 :	1	1	1	1	0	0
1 :	1	0	0	0	0	0
2 :	0	0	0	0	0	1

→ Take 2 clock edges to come to 000, continuing - -

3 :	0	0	1	0	1	0
4 :	0	1	0	0	1	1
5 :	0	1	1	1	0	0
6 :	1	0	0	0	0	0
7 :	0	0	0			

4) States between 000 and 000 are 001, 010, 011, 100